Tools: SQL Server

MARKETING CAMPAIGN PERFORMANCE ANALYSIS USING SQL

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BUSINESS QUESTIONS

- Which campaigns generate the highest profitability?
- Which marketing channels are the most effective?
- Which campaigns require further evaluation?
- How can marketing budgets be optimized?

BUSINESS UNDERSTANDING

The company runs multiple marketing campaigns to increase sales of healthcare products. However, several key questions remain unanswered:

- Which campaigns generate the highest profitability?
- Which marketing channels are the most effective?
- Which campaigns contribute to budget inefficiencies?

BUSINESS OBJECTIVES

- Measure marketing campaign performance.
- Identify the marketing channels with the highest conversion rates.
- Identify underperforming campaigns that require further evaluation.
- Provide data-driven recommendations to optimize marketing spending.

EXECUTIVE SUMMARY

This project aims to evaluate marketing campaign performance using SQL to identify the most effective campaigns and channels, measure profitability, and generate data-driven business recommendations.



Best Campaign

Flash Sale P3K
Liburan
ROAS = 16.49



Best Channel

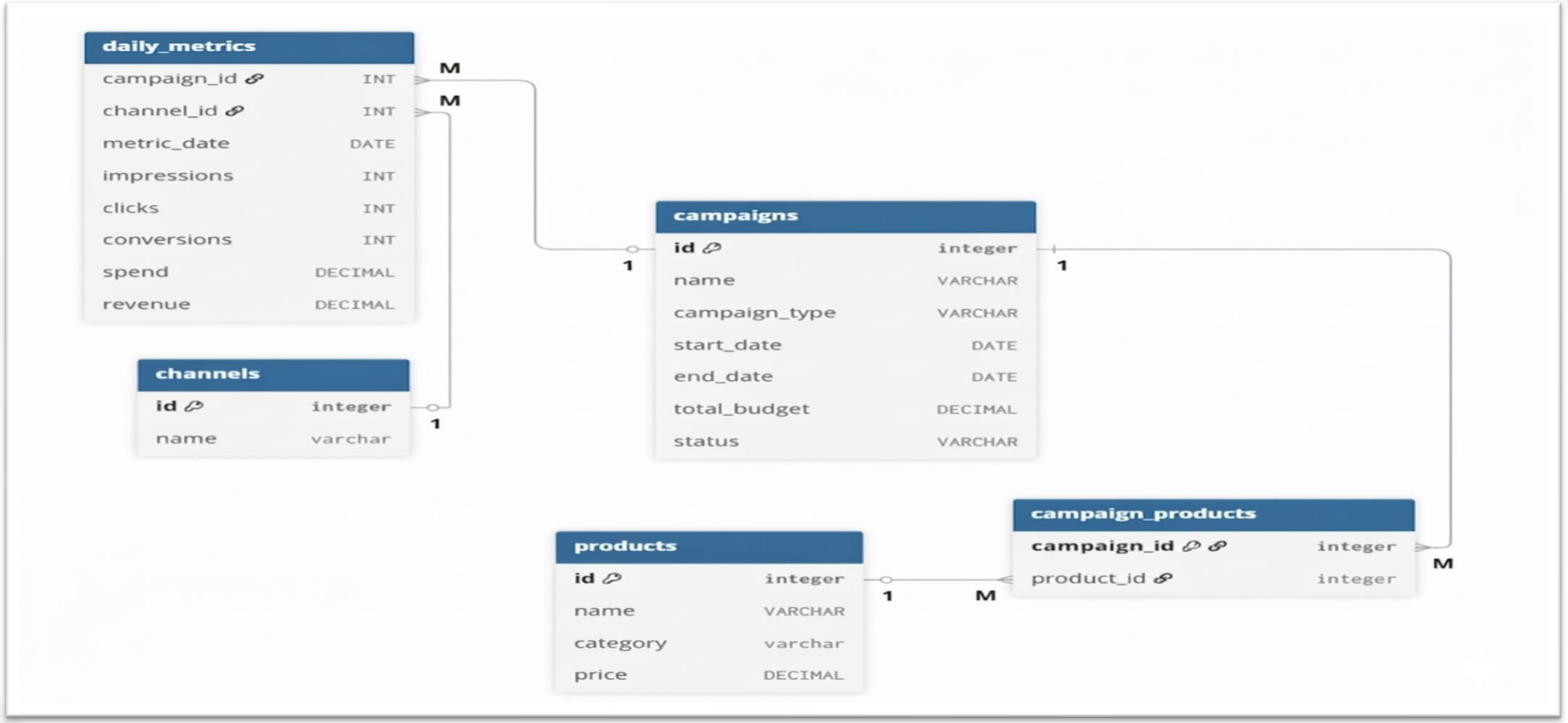
CRM & WhatsApp
CTR = 10.4%
CVR = 8.33%



Worst Campaign

Launch Acne Care
Gel Klinis
ROAS = 0.64

ENTITY-RELATIONSHIP DIAGRAM (ERD)



DATA MODEL

Dataset Overview

Table	Description
daily_metrics	Daily performance metrics by campaign and channel.
campaigns	Marketing campaign master data.
channels	Digital marketing channel master data.
products	Product master data.
campaign_products	Mapping table between campaigns and products.

Analytical Scope

- Campaign Analysis
- Channel Analysis
- ROAS Analysis
- Profitability Analysis
- Trend Analysis

DATA QUALITY

Business Question

- Is the dataset reliable enough for analysis?

SQL Query

```sql
/**============================================================================
 * TAHAP 1: DATA QUALITY CHECK & PROFILING (Tabel: daily_metrics)
 * ============================================================================**/
SELECT
    -- Menghitung total seluruh catatan data dalam tabel
    COUNT(*) AS total_rows,

    -- Memeriksa jumlah ID unik untuk mendeteksi potensi duplikasi data
    COUNT(DISTINCT id) AS unique_rows,

    -- Menghitung jumlah data kosong (NULL) pada biaya iklan untuk antisipasi bias analisis ROI
    SUM(CASE WHEN spend IS NULL THEN 1 ELSE 0 END) AS null_spend,

    -- Menghitung jumlah data kosong (NULL) pada pendapatan untuk memastikan keakuratan pelaporan ROAS
    SUM(CASE WHEN revenue IS NULL THEN 1 ELSE 0 END) AS null_revenue
FROM daily_metrics;
```

Insight:

- No duplicate records were identified.
- No missing values were found in key metrics.
- The dataset is suitable for marketing performance analysis.

Result

Metric	Value
Total Rows	120
Unique Rows	120
Duplicate Rows	0
Missing Values	0

CAMPAIGN PERFORMANCE ANALYSIS

Business Question

- Which campaigns deliver the best performance?

SQL Query

```sql
/**============================================================
 * METRIK BISNIS: Top 3 Campaign Efisiensi Tertinggi (ROAS)
 * ============================================================
-- CTE Agregasi total pengeluaran & pendapatan per campaign
WITH campaign_performance AS (
    SELECT
        c.name AS campaign_name,
        SUM(dm.spend) AS total_spend,
        SUM(dm.revenue) AS total_revenue
    FROM campaigns c
    JOIN daily_metrics dm ON c.id = dm.campaign_id
    GROUP BY c.name
)
SELECT TOP 3
    campaign_name,
    -- Menghitung ROAS
    ROUND(total_revenue / NULLIF(total_spend, 0), 2) AS roas
FROM campaign_performance
ORDER BY roas DESC
```

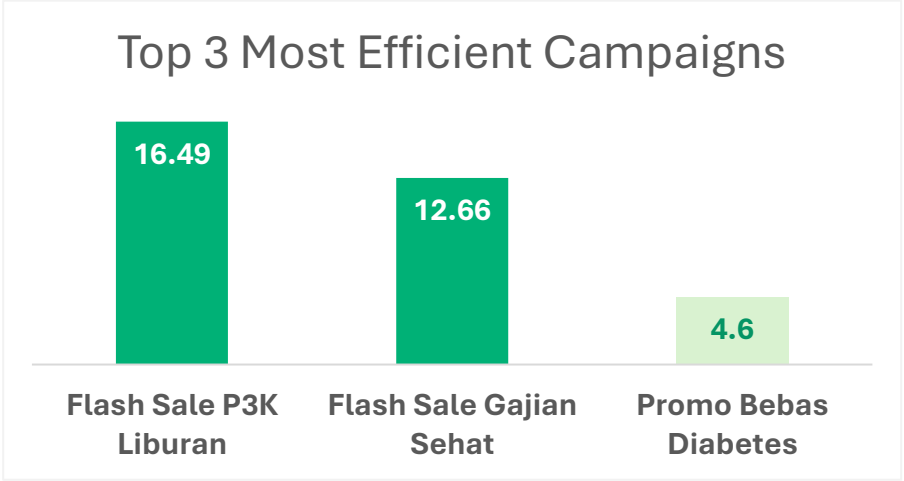
Insight:

- The two Flash Sale campaigns generated the highest ROAS compared to other campaigns.

Business Interpretation:

- Urgency-based promotional strategies have proven effective in improving marketing cost efficiency.

Result



CAMPAIGN TYPE ANALYSIS

Business Question

- Which campaign type is the most effective?

SQL Query

```sql
-- ============================================
-- PERFORMA FINANSIAL BERDASARKAN TIPE CAMPAIGN
-- ============================================
SELECT
    c.campaign_type,
    COUNT(DISTINCT c.id) AS jumlah_campaign,
    SUM(dm.spend) AS total_spend,
    SUM(dm.revenue) AS total_revenue,
    ROUND(
        SUM(dm.revenue) /
        NULLIF(SUM(dm.spend), 0), 2
    ) AS roas
FROM campaigns c
JOIN daily_metrics dm
    ON c.id = dm.campaign_id
GROUP BY c.campaign_type
ORDER BY roas DESC;
```

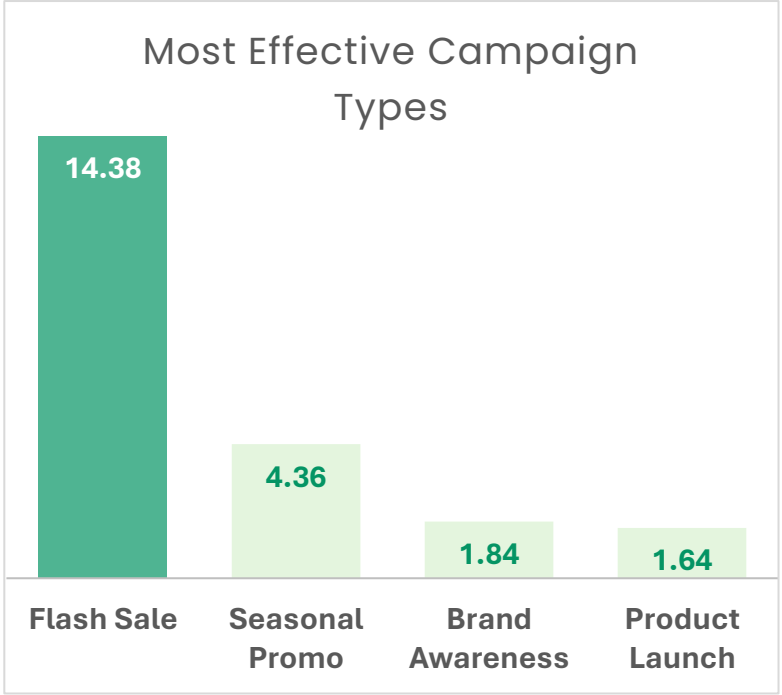
Insight:

- Flash Sale campaigns achieved a ROAS of 14.38x, which is 8.8 times higher than Product Launch campaigns (1.64x).

Business Impact:

- The Flash Sale strategy can serve as a model for future marketing campaigns.

Result



CHANNEL PERFORMANCE ANALYSIS

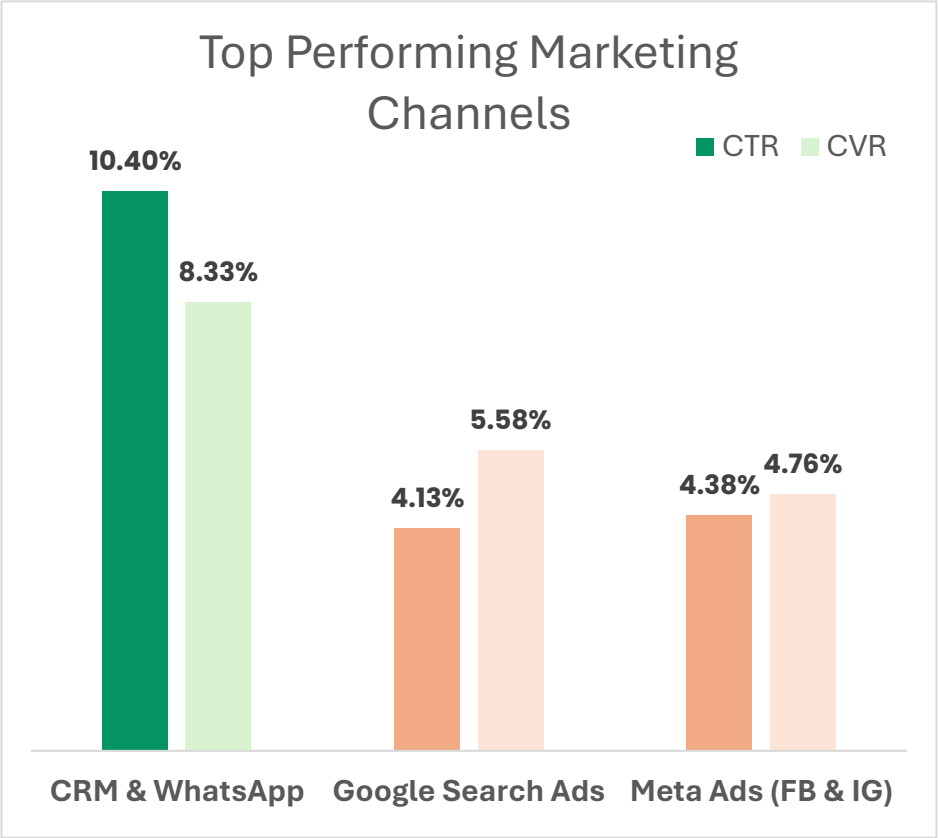
Business Question

Which marketing channels deliver the best performance?

SQL Query

```sql
-- ============================================================
-- PERFORMA IKLAN BERDASARKAN PERSENTASE KONVERSI (CONVERSION RATE)
-- ============================================================
SELECT
    c.name AS channel_name,
    SUM(dm.spend) AS total_spend,
    SUM(dm.revenue) AS total_revenue,
    ROUND(
        SUM(dm.conversions) * 100.0 /
        NULLIF(SUM(dm.clicks), 0), 2
    ) AS conversion_rate
FROM channels c
JOIN daily_metrics dm
    ON c.id = dm.channel_id
GROUP BY c.name
ORDER BY conversion_rate DESC;
```

Result



CHANNEL PERFORMANCE ANALYSIS

Insight:

- CRM & WhatsApp achieved the highest CTR (10.40%) and CVR (8.33%), making it the most effective marketing channel for driving engagement and conversions.

Business Interpretation:

- Direct communication channels, such as CRM & WhatsApp, generate the highest-quality traffic and should be prioritized in marketing budget allocation.
  In contrast, KOL Medical Review requires further evaluation due to its low conversion rate.

PROFITABILITY ANALYSIS

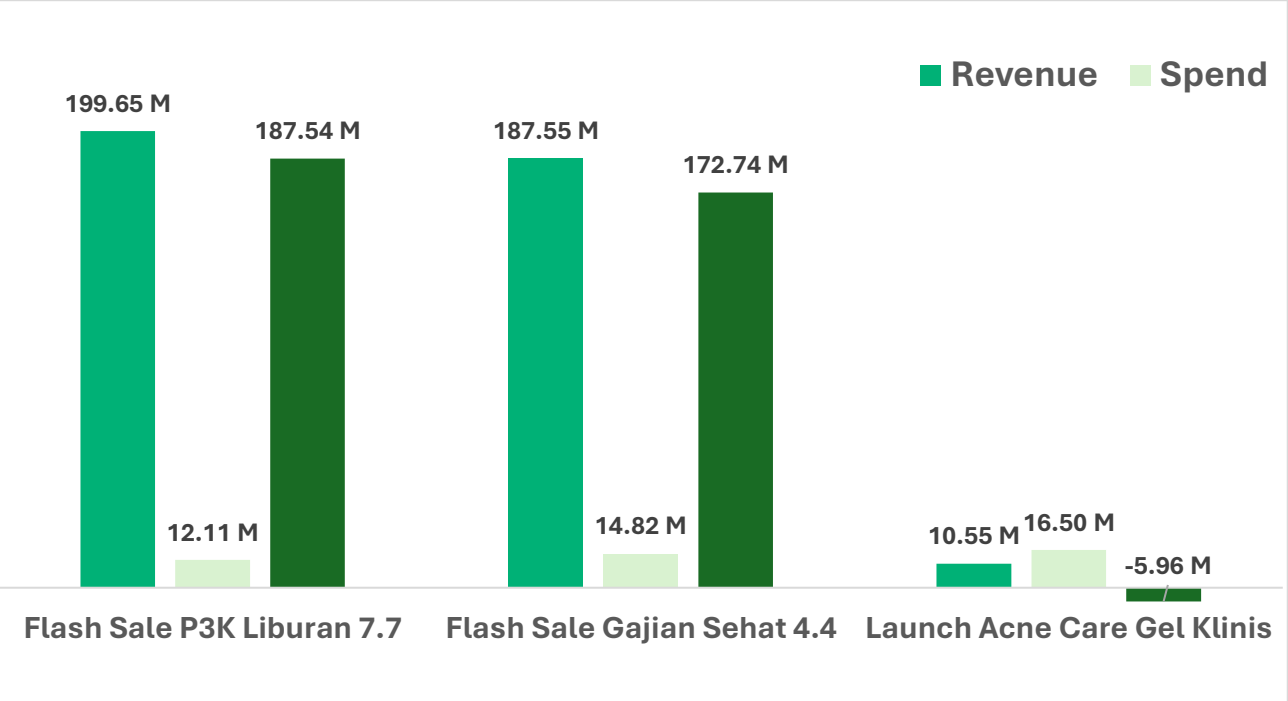
Business Question

Does higher revenue always lead to higher profit?

SQL Query

```sql
-- ============================================
-- ANALISIS PROFIT BERDASARKAN NAMA CAMPAIGN
-- ============================================
SELECT
    c.name AS campaign_name,
    SUM(dm.revenue) AS total_revenue,
    SUM(dm.spend) AS total_spend,
    SUM(dm.revenue) - SUM(dm.spend) AS total_profit
FROM campaigns c
JOIN daily_metrics dm
    ON c.id = dm.campaign_id
GROUP BY c.name
ORDER BY total_revenue DESC;
```

Result



PROFITABILITY ANALYSIS

Insight:

- Higher revenue does not always result in higher profit; campaign effectiveness should be evaluated based on both profit and ROAS.

- Flash Sale P3K Liburan generated the highest profit of Rp187.54 million, while Launch Acne Care Gel Klinis incurred a loss of Rp5.95 million.

Business Impact:

- Using profit and ROAS as primary KPIs can help optimize marketing budget allocation and improve overall profitability.

ROOT CAUSE ANALYSIS

Campaign Under Review

Launch Acne Care Gel Klinis

KPI	Value	Interpretation
CTR	2.42%	Low audience engagement
CVR	1.71%	Low conversion effectiveness
ROAS	0.64x	Revenue did not cover marketing costs

Key Finding

- This campaign failed to generate sufficient revenue to cover marketing costs, resulting in negative profitability.

ROOT CAUSE ANALYSIS

Root Cause

- Low CTR (2.42%)

  The campaign struggled to capture audience interest and encourage clicks.
- Low CVR (1.71%)

  Only a small proportion of visitors completed the desired action or made a purchase.
- Audience Misalignment

  The target audience may not align with the product positioning or campaign messaging.

Business Impact

- For every Rp1 spent on marketing, the campaign generated only Rp0.64 in revenue. This campaign reduced overall profitability.

WEEKLY GROWTH ANALYSIS

Business Question

- Is revenue growth stable over time?

Key Findings

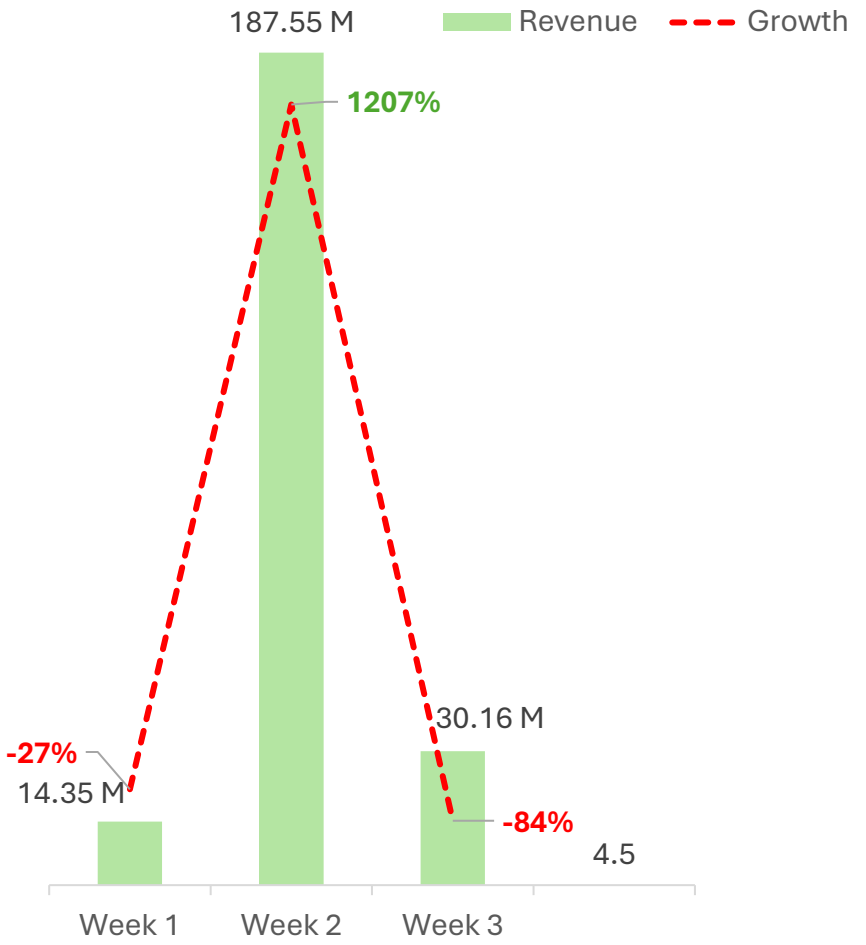
- Revenue increased by **1,207%** during the week of March 24.
- However, it declined by **84%** in the following week.

Business Impact

- Revenue growth remains highly dependent on specific campaigns.

Business Risk

- The business has not yet established a consistent and sustainable growth pattern.



BUSINESS RECOMMENDATIONS & CONCLUSION

Recommendations

High Priority

- Prioritize budget allocation for CRM & WhatsApp.
- Replicate successful Flash Sale strategies.

Medium Priority

- Reassess Product Launch campaigns.
- Improve audience targeting.

Long-Term Strategy

- Monitor ROAS regularly.
- Conduct A/B testing for future campaigns.

Conclusion

The SQL analysis successfully identified the highest-performing campaigns and channels, uncovered sources of marketing inefficiency, and generated actionable recommendations to improve budget allocation effectiveness and overall business profitability.

“Turning complex data into clear business direction.”

INTERESTED IN WORKING TOGETHER?

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THANK YOU FOR YOUR TIME